CSE211: Computer Organization and Design

UNIT 4: Introduction to HDLs in Digital Systems

Topics:

History, Evolution, Typical HDL-based Design Flow (Design, Simulation, Synthesis, Verification), Comparison with Software Programming Languages, Verilog.

Multiple Choice Quepstions

1. Which keyword in Verilog is used to define a combinational logic block that re-evaluates whenever any input changes?
   a. always @(posedge clk)
   b. **always @(*)**
   c. initial
   d. forever
2. What is the difference between 'reg' and 'wire' in Verilog?
   a. reg is for clocked logic only; wire is for asynchronous logic
   b. wire holds a value between assignments; reg does not
   c. **reg can hold a value between assignments; wire must be continuously driven**
   d. There is no difference — both are interchangeable
3. Which Verilog construct is used to model a flip-flop triggered on the positive edge of a clock?
   a. always @(*)
   b. **always @(posedge clk)**
   c. assign
   d. initial begin
4. In Verilog, the 'assign' statement is used for:
   a. Sequential logic only
   b. **Continuous assignment to wire types**
   c. Initialization of reg variables
   d. Defining test benches
5. Which of the following is a correct 4-bit binary literal in Verilog?
   a. **4'b1010**
   b. B'1010
   c. 4h'A
   d. binary(1010)
6. What does the '$finish' system task do in a Verilog simulation?
   a. Resets all signals to zero
   b. **Terminates the simulation**

c. Writes output to a file

d. Starts a new simulation run

7. Which operator in Verilog performs bitwise XOR?

a. ~

b. ^

c. &&

d. |

8. In a Verilog module, 'input', 'output', and 'inout' are used to declare:

a. Local variables

b. **Module port directions**

c. Simulation parameters

d. Compiler directives

9. Which step in the HDL-based design flow converts HDL code into a gate-level netlist?

a. Simulation

b. **Synthesis**

c. Verification

d. Placement & Routing

10. What is the primary purpose of RTL simulation in the HDL design flow?

a. To map the design to physical gates

b. **To verify functional correctness before synthesis**

c. To measure power consumption

d. To generate layout files

11. Verilog was originally developed by which company?

a. Mentor Graphics

b. **Cadence / Gateway Design Automation**

c. Synopsys

d. Intel

12. Which IEEE standard number corresponds to the original Verilog HDL?

a. IEEE 1076

b. **IEEE 1364**

c. IEEE 1800

d. IEEE 1149

13. What is the value of 'z' in Verilog?

a. Logic 0

b. Logic 1

c. **High impedance (tri-state)**

d. Unknown/undefined

14. Which Verilog module instantiation syntax correctly connects ports by name?

a. mymod u1(a, b, c);

b. **mymod u1(.out(c), .in1(a), .in2(b));**

c. mymod u1{a=>in1, b=>in2};

d. instance mymod(a,b,c);

15. During synthesis, 'timing constraints' are used to:

a. Set the simulation time limit

b. **Guide the synthesizer to meet target clock frequency and setup/hold times**

c. Define the number of test vectors

d. Specify power supply voltage